T-SCALED REALIZATION OF p-ADIC HAHN SERIES

by

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1. Preliminaries on Hahn series

To make this article self-contained, we briefly recall some basic facts about Hahn series.

Definition 1.1 **([Poo93, Section 3]).** — *Let R be a commutative ring and G be an ordered group.*

1. *For any $f \in \mathrm{Hom}_{\mathrm{Set}}(G, R)$, we define the* ***support*** *of f to be*
$$\mathrm{Supp}(f) = \{g \in G \colon f(g) \neq 0\}.$$
2. *Define the set of* ***Hahn series*** *over R with value group G to be*
$$R(\!(G)\!) \coloneqq \{f \in \mathrm{Hom}_{\mathrm{Set}}(G, R) \colon \mathrm{Supp}(f) \text{ is well-ordered}\}.$$
By introducing a formal variable t, elements in $R(\!(G)\!)$ will also be written as $\sum_{g\in G} r_g t^g$, where $r_g \in R$ for all $g \in G$.

Proposition 1.2 **([Poo93, Lemma 1, Corollary 2]).** — *Let R be a commutative ring and G be an ordered group.*

1. *With identity $1 \cdot t^0$ and addition as well as multiplication given by*
$$\sum_{g\in G} a_g t^g + \sum_{g\in G} b_g t^g \coloneqq \sum_{g\in G} (a_g + b_g) t^g, \ \sum_{g\in G} a_g t^g \cdot \sum_{g\in G} b_g t^g \coloneqq \sum_{g\in G} \left(\sum_{h\in G} a_h b_{g-h} \right) t^g,$$
$R(\!(G)\!)$ forms a commutative ring.
2. *If R is a field, then so is $R(\!(G)\!)$. Moreover, with the map*
$$v \colon R(\!(G)\!) \longrightarrow G \cup \{\infty\}, \ f \longmapsto \begin{cases} \min \mathrm{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases},$$
$R(\!(G)\!)$ becomes a valued field with value group G and residue field R.

Note that $\mathrm{char}\, R(\!(G)\!) = \mathrm{char}\, R$, we call $R(\!(G)\!)$ the **equal-characteristic field of Hahn series** over R with value group G, also denoted as $R(\!(t^G)\!)$ with respect to the formal variable t.

Proposition 1.3 **([Poo93, Proposition 3, Corollary 3, Proposition 5])**
Let k be a perfect field of characteristic p and G be an ordered group containing $\mathbb{Z}$ as a subgroup. Besides that, let

$$\mathcal{N} := \left\{ \sum_{g\in G} r_g t^g \in W(k)(\!(t^G)\!) \colon \text{ for every } g \in G,\ \sum_{n\in\mathbb{Z}} r_{g+n} p^n = 0 \right\},$$

where $W(k)$ is the ring of Witt vectors of k. We call elements in $\mathcal{N}$ the ***null series*** *of $W(k)(\!(t^G)\!)$. Then*

1. *$\mathcal{N}$ is a maximal ideal of $W(k)(\!(t^G)\!)$, which makes $W(k)(\!(p^G)\!) := W(k)(\!(t^G)\!)/\mathcal{N}$ a field*[(1)]*, called the* ***p-adic field of Hahn series****.*
2. *Every element in $W(k)(\!(p^G)\!)$ can be uniquely written as*

$$\sum_{g\in G} [r_g] p^g,$$

where $r_g \in k$ for all $g \in G$ and $[\cdot] \colon k \longrightarrow W(k)$ is the Teichmüller lift. We call this the ***standard expansion*** *of the element.*

3. *For $f = \sum_{g\in G} [r_g] p^g$, define the* ***support*** *of f to be*

$$\operatorname{Supp}(f) = \{g \in G \colon r_g \neq 0\}.$$

Then the map

$$v \colon W(k)(\!(G)\!)/\mathcal{N} \longrightarrow G \cup \{\infty\},\ f \mapsto \begin{cases} \min \operatorname{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases}$$

makes $W(k)(\!(G)\!)/\mathcal{N}$ a mixed-characteristic valued field with value group G and residue field k.

The most important property of the field of Hahn series is the following:

Theorem 1.4 **(cf. [Poo93, Theorem 1, Corollary 4, Corollary 6])**
Let F be an equal-characteristic (resp. mixed-characteristic) valued field with divisible value group G and algebraically closed residue field k. Then the equal-characteristic (resp. p-adic) field of Hahn series $k(\!(t^G)\!)$ (resp. $W(k)(\!(p^G)\!)$) is the unique (up to isomorphisms of valued field) minimal spherically complete extension of F. Moreover, it is algebraically closed and complete.

The following examples are the fields of Hahn series used in this article:

Example 1.5. — *Let $F = \overline{\mathbf{F}}_p(\!(t)\!)$ (resp. $\breve{\mathbf{Q}}_p = W(\overline{\mathbf{F}}_p)[p^{-1}]$), which has value group $\mathbf{Q}$ and residue field $\overline{\mathbf{F}}_p$. Then the field of equal-characteristic (resp. p-adic) Hahn series $\overline{\mathbf{F}}_p(\!(t^{\mathbf{Q}})\!)$ (resp. $\mathscr{O}_{\breve{\mathbf{Q}}_p}(\!(p^{\mathbf{Q}})\!) = W(\overline{\mathbf{F}}_p)(\!(p^{\mathbf{Q}})\!)$) is the spherical completion of F with residue field and value group unchanged, which is algebraically closed and complete.*

Remark 1.6. — *To simplify the statement, we will simply call $\mathscr{O}_{\breve{\mathbf{Q}}_p}(\!(p^{\mathbf{Q}})\!)$ the p-adic Hahn series without specifying the residue field and value group.*

2. T-scaled realization of $\mathbf{L}_p$

During this section, T will be an positive integer.

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(1) Intuitively speaking, $W(k)(\!(p^G)\!)$ is obtained by replacing the formal variable t of elements in $W(k)(\!(t^G)\!)$ by the prime p.

Lemma 2.1. — *One has* $\left[\breve{\mathbf{Q}}_p(p^{1/T})\colon \breve{\mathbf{Q}}_p\right] = T$. *In particular,* $1, p^{1/T}, \cdots, p^{(T-1)/T}$ *form a basis of* $\breve{\mathbf{Q}}_p(p^{1/T})$ *over* $\breve{\mathbf{Q}}_p$.

Proof. — Since $\breve{\mathbf{Q}}_p$ is a discrete valuation field and $p^{1/T}$ is a root of the Eisenstein polynomial $X^T - p$, the result follows from the Eisenstein criterion. □

Lemma 2.2. — *Every element of* $W(\overline{\mathbf{F}}_p)[p^{1/T}]$ *can be uniquely written as* $\sum_{k=0}^{\infty}[c_k]p^{\frac{k}{T}}$.

Proof. — This is a simple corollary of the fact that every element in $W(\overline{\mathbf{F}}_p)$ can be uniquely written as $\sum_{k=0}^{\infty}[c_k]p^k$. The proof may also need Lemma 2.1. □

Remark 2.3. — $W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ *is a subfield of* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ *via the natural inclusion* $W(\overline{\mathbf{F}}_p) \subset W(\overline{\mathbf{F}}_p)[p^{1/T}]$.

Definition 2.4. — *Let* $\mathcal{N}_T$ *be the set of elements* $\sum_{q\in\mathbf{Q}} c_q t^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ *that for any* $q \in \mathbf{Q}$,

$$\sum_{n\in\mathbf{Z}} c_{q+\frac{n}{T}} p^{\frac{n}{T}} = 0.$$

We call these elements the T***-null-series*** *in* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.

Remark 2.5. — *Obviously when T=1,* $\mathcal{N}_T$ *is the null-series in* $W(\overline{\mathbf{F}}_p)(\!(p^{\mathbf{Q}})\!)$ *defined in Proposition 1.3.*

Lemma 2.6. —

1. $\mathcal{N}_T$ *is an ideal of* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.
2. *For every element* f *of* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, *there exists a unique element* $g = \sum_{q\in\mathbf{Q}}[g(q)]t^{\mathbf{Q}}$ *in* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ *such that* $f - g \in \mathcal{N}_T$.
3. $\mathcal{N}_T$ *is a maximal ideal of* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, *which makes* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$ *a field.*

Proof. — These three statements are generalizations of [Poo93, Proposition 3], [Poo93, Proposition 4] and [Poo93, Corollary 3] respectively, and the proofs are every similar. □

Remark 2.7. — *By (2) of this lemma, we will formally write elements of* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$ *as* $\sum_{q\in\mathbf{Q}}[g(q)]p^q$.

Lemma 2.8. — *One has* $\mathcal{N}_T \cap W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) = \mathcal{N}$.

Proof. — Let $f = \sum_{q\in\mathbf{Q}}[f(q)]t^q \in \mathcal{N}$, then for any $q \in \mathbf{Q}$,

$$\sum_{n\in\mathbf{Z}}[f(q+n)]p^n = 0. \tag{a}$$

Notice that for any $q \in \mathbf{Q}$, one has

$$\begin{aligned}\sum_{n\in\mathbf{Z}}\Big[f\Big(q+\frac{n}{T}\Big)\Big]p^{\frac{n}{T}} &= \sum_{u=0}^{T-1}\sum_{\substack{n\in\mathbf{Z}\\ n\equiv u \pmod T}}\Big[f\Big(q+\frac{n}{T}\Big)\Big]p^{\frac{n}{T}}\\ &= \sum_{u=0}^{T-1}p^{\frac{u}{T}}\sum_{n\in\mathbf{Z}}\Big[f\Big(\Big(q+\frac{u}{T}\Big)+n\Big)\Big]p^n.\end{aligned}$$

By applying (a) to $q + \frac{u}{T}$ for $u = 0, 1, \cdots, T-1$, one has $\sum_{n\in\mathbf{Z}}\big[f\big(q+\frac{n}{T}\big)\big]p^{\frac{n}{T}} = 0$. As a result, $f \in \mathcal{N}_T$.

Conversely, let $g = \sum_{q\in\mathbf{Q}}[g(q)]t^q \in \mathcal{N}_T$. Then for any $q \in \mathbf{Q}$, one has

$$\sum_{n\in\mathbf{Z}}\Big[g\Big(q+\frac{n}{T}\Big)\Big]p^{\frac{n}{T}} = \sum_{u=0}^{T-1} p^{\frac{u}{T}} \sum_{n\in\mathbf{Z}}\Big[g\Big(\Big(q+\frac{u}{T}\Big)+n\Big)\Big]p^n = 0.$$

By Lemma 2.1, for any $u = 0, 1, \cdots, T-1$, one has $\sum_{n\in\mathbf{Z}}\big[g\big((q+\frac{u}{T})+n\big)\big]p^n = 0$. The result follows from taking $u = 0$. □

Lemma 2.9. — *One has* $W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) + \mathcal{N}_T = W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.

Proof. — Take $f = \sum_{q\in\mathbf{Q}} c_q T^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, where

$$c_q = \sum_{n=0}^{\infty}[c_{q,n}]p^{\frac{n}{T}} \in W(\overline{\mathbf{F}}_p)[p^{1/T}].$$

Then one has $f = f_0 + p^{\frac{1}{T}} f_1 + \cdots + p^{\frac{T-1}{T}} f_{T-1}$, where

$$f_u = \sum_{t\in\mathbf{Q}}\left(\sum_{\substack{n\in\mathbf{N} \\ n\equiv u \pmod{T}}}[c_{q,n}]p^{\frac{n-u}{T}}\right)t^q \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})),\ u = 0, 1, \cdots, T-1.$$

It suffices to show that $p^{\frac{u}{T}} f_u$ belongs to $W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) + \mathcal{N}_T$ for $u = 0, 1, \cdots, T-1$. This is clear if we write f_u as $f_u = (p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}})f_u + t^{\frac{u}{T}} f_u$, where $t^{\frac{u}{T}} f_u \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ and $(p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}})f_u$ is a T-null-series. □

Proposition 2.10. — *One has isomorphism of valued fields*

$$\sigma\colon \mathbf{L}_p \xrightarrow{\cong} W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T.$$

Moreover, the following diagram commutes:

$$\text{(b)}\qquad \begin{array}{ccc} W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) & \xhookrightarrow{\ \iota\ } & W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}})) \\ \big\downarrow{\scriptstyle \pi} & & \big\downarrow{\scriptstyle \pi_T} \\ \mathbf{L}_p & \xrightarrow[\sigma]{\cong} & W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T \end{array}.$$

Proof. — This is the direct consequence of the following standard fact in commutative algebra: given domains $R_1 \subset R_2$ and an ideal I of R_2, if $R_1 + I = R_2$, then $R_1/(I \cap R_1) \cong R_2/I$, where the isomorphism is induced by the inclusion $R_1 \subset R_2$. □

Remark 2.11. — *For any element* $f = \sum_{q\in\mathbf{Q}}[f(q)]p^q \in \mathbf{L}_p$, *its lift* $\widehat{f} = \sum_{q\in\mathbf{Q}}[f(q)]t^q \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ *embeds into* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ *and then can be projected to the element in* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$, *which can also be formally written as* $\sum_{q\in\mathbf{Q}}[f(q)]p^q$. *The commutative diagram in the above proposition above implies that the isomorphism between* $\mathbf{L}_p$ *and* $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$ *is given by the formula*

$$\sum_{q\in\mathbf{Q}}[f(q)]p^q \in \mathbf{L}_p \longmapsto \sum_{q\in\mathbf{Q}}[f(q)]p^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T.$$

References

[Poo93] B. Poonen. "Maximally Complete Fields". In: *L'Enseignement Mathématique* 39.1-2 (1993), p. 87. DOI: 10/kqcb.

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